Mass Deficit Nodes and Curvature Extremum Duality

Void Nodes as Geometrically Complementary Organizers of Spacetime

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Abstract

Observations show that the universe is dominated by expansive, underdense regions by volume, while mass is confined to compact structures such as galaxies, clusters, and black holes. In this work we identify Mass Deficit Nodes (MDNs) as local minima of spacetime curvature that organize expansion-dominated cosmic voids. Additionally, we identify MDNs as geometrically complementary counterparts to mass-concentration nodes that organize matter-dominated structures. Together, mass-concentration nodes and mass deficit nodes define a dual organizing structure of spacetime curvature extrema, which we refer to as Curvature Extremum Duality.

Within this framework, large cosmic voids are not merely the absence of matter but the extended manifestations of curvature minima that actively shape spacetime geometry. A primary and observationally accessible consequence of this structure is a path-integrated geometric contribution to cosmological redshift, accumulated predominantly during photon propagation through void-organized geometry. Early-universe physics is preserved, while late-time predictions arise naturally from observed large-scale structure. These include correlations between redshift residuals and line-of-sight void geometry, redshift excesses behind large supervoids at fixed independent distance, and systematic implications for locally inferred expansion rates.

The framework is fully consistent with general relativity and relies only on empirically established features of the cosmic web.

1. Introduction and Motivation

Large-scale structure surveys reveal a pronounced asymmetry between the distribution of mass and the distribution of volume in the universe. Matter resides primarily in compact structures such as galaxies, clusters, and black holes, while the overwhelming fraction of cosmic volume consists of expansive underdense regions commonly referred to as voids. Photons propagating across cosmological distances therefore spend the majority of their path length traversing expansion-dominated environments rather than matter-dominated ones.

Standard cosmology models the universe as approximately homogeneous on sufficiently large scales, treating voids and filaments as perturbations around a smooth background expansion.

This approximation has been extraordinarily successful. However, the volumetric dominance of void regions invites closer scrutiny of the geometric role these regions play in shaping spacetime and influencing observable quantities along null paths.

There is direct empirical evidence that void structure affects photon energy. The Integrated Sachs–Wolfe (ISW) effect demonstrates that photons traversing large underdense regions experience measurable energy shifts associated with gravitational geometry. The present work builds on this observation by stepping back one level conceptually: instead of treating void effects as small corrections to a homogeneous background, we ask whether the geometric structures that organize voids themselves (curvature minima) constitute fundamental spacetime features with systematic observational consequences.

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2. Curvature Extremum Duality: Conceptual Framework

2.1 Organizing nodes of spacetime geometry

General relativity permits spacetime curvature to organize structure. Compact mass concentrations generate curvature maxima, and familiar astrophysical objects such as black holes act as organizing centers around which matter clusters, null congruences converge, and causal structure becomes constrained.

In this work we identify the complementary possibility: curvature minima that organize surrounding spacetime into expansion-dominated regions. We refer to these organizing minima as Mass Deficit Nodes (MDNs). For descriptive clarity, we will occasionally use the colloquial term void node in the context of a node that approaches an absolute minimum.

We emphasize that an MDN is a geometric node, not a material object. It denotes the organizing curvature minimum around which a large cosmic void is structured, in direct analogy with a black hole as the curvature maximum organizing a galaxy or cluster.

2.2 Curvature Extremum Duality

Taken together, mass-concentration nodes (curvature maxima) and mass-deficit nodes (curvature minima) define a dual organizing structure of spacetime curvature extrema, which we refer to formally as Curvature Extremum Duality.

This duality does not imply geometric inversion, metric sign reversal, or dynamical equivalence. Rather, it asserts that spacetime organizes around both extrema of curvature, with complementary, but not identical, physical manifestations:

- curvature maxima organize matter via convergence and accretion,
- curvature minima organize spacetime via divergence and evacuation.

Some phenomena admit natural complements across these extrema while others do not. These asymmetries encode real physics rather than weaknesses of the framework.

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3. Mass Deficit Nodes as Geometric Objects

A Mass Deficit Node (MDN) is defined geometrically as a local minimum of spacetime curvature within the cosmic web, standing in complementary relation to a mass-concentration node (a local curvature maximum).

An MDN organizes surrounding spacetime into an extended underdense region (a cosmic void). Matter evacuation and void persistence follow from the geometry rather than define it. While density undercontrast provides an observational tracer of MDN-organized regions, it is the curvature minimum that defines the node itself.

Not all underdense regions qualify as MDNs. Many voids may be sub-critical or transient. The MDN concept identifies the organizing curvature minima associated with the largest, most persistent void structures.

MDNs do not arise from local collapse. Instead, they represent the geometric endpoint of an expansion-driven evacuation process: initially underdense regions experience reduced gravitational resistance to expansion, expand faster than average, lose matter to surrounding structures, and deepen their curvature minima. This feedback naturally drives such regions toward stable geometric basins.

In this hierarchy:

- the MDN (void node) is the organizing curvature minimum (complementary to a black hole),
- the void is the extended structure organized around the node (analogous to a galaxy or halo).

While the MDN refers to the ultimate curvature minimum organizing a void, the associated geometry is not necessarily point-like. Curvature minima may be distributed through the void interior in a basin-like form with the MDN denoting the organizing minimum rather than the full spatial extent or internal structure of the void.

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4. A Primary Observable Consequence: Path-Integrated Geometric Redshift

4.1 Dominance of void-organized geometry along null paths

Photon propagation along cosmological null geodesics alternates between matter-dominated regions and underdense regions. Although matter-dominated regions exert strong local influence, they occupy a small fraction of cosmic volume. Over large distances, photon propagation is therefore dominated by traversal through void regions organized around MDNs, which control the majority of affine path length.

Within these expansion-dominated regions, null congruences diverge and spacetime intervals stretch. This produces cumulative, adiabatic wavelength stretching that preserves phase-space density and spectral form. Unlike tired-light mechanisms, this effect involves no photon-medium interaction, no scattering, and no image degradation.

4.2 Redshift as an affine path-length measure

Path-Integrated Geometric Redshift Principle.
In a void-dominated universe, observed cosmological redshift encodes the affine path length of photon propagation through MDN-organized spacetime geometry rather than serving solely as a proxy for kinematic recession.

In general relativity, coordinate distance is not physically invariant; physical distance is determined by the metric and by the path taken. Mass-concentration nodes compress effective photon path lengths via convergence and lensing. MDNs, by contrast, inflate effective affine path lengths via divergence of null congruences.

Potential-based redshift effects such as ISW and Rees–Sciama rely on time-varying gravitational potentials and exhibit near-cancellation for static structures. MDN geometric redshift instead reflects cumulative adiabatic stretching driven by geometry itself. Because the stretched waveform is carried forward, no reciprocal “exit” process reverses the effect.

4.3 Observational consequences

Three testable consequences follow naturally:

1. Redshift residuals at fixed distance.
   Objects viewed through large MDN-organized void regions should exhibit excess redshift relative to objects at equal independently measured distance viewed along less void-dominated lines of sight.

2. Correlation with void geometry.
   Redshift residuals should correlate with the integrated void geometry organized around MDNs along the line of sight.

3. Implications for locally inferred expansion rates.
   If redshift includes a geometric contribution from MDN-organized path-length inflation, local measurements that attribute all redshift to recession velocity will systematically overestimate expansion rates, contributing a geometric component to the observed Hubble tension.

4.4 Blueshifted galaxies as a consistency check

Nearby gravitationally bound systems provide an internal check. Galaxies within the Local Group exhibit blueshift due to actual kinematic approach, as their light traverses minimal void-organized geometry. At larger distances, accumulated MDN geometric redshift overwhelms any plausible kinematic blueshift, producing the observed transition to universal redshift without ad hoc exceptions.

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5. Implications for Background Radiation

The framework does not challenge the early-universe origin of the cosmic microwave background (CMB) or the physics of recombination. Instead, it examines how late-time propagation through MDN-organized geometry shapes the observed radiation.

CMB photons traverse billions of years of expansion-dominated spacetime organized around curvature minima. Adiabatic geometric redshifting preserves the Planckian spectrum while contributing to cumulative wavelength stretching. The ISW effect already demonstrates correlation between large voids and CMB temperature fluctuations; the MDN framework interprets this as a systematic late-time geometric influence layered atop early-universe physics.

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6. Limitations and Scope

This work is restricted to late-time spacetime geometry and photon propagation. It does not modify primordial nucleosynthesis, recombination, or early-universe thermodynamics. Light-element abundances remain intact constraints.

A full metric-level derivation of redshift accumulation in explicitly MDN-structured spacetime remains to be developed. Quantum and horizon-level implications arise only in extreme curvature-minimum regimes and are deferred to future work.

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7. Conclusion

We have identified Mass Deficit Nodes (void nodes) as geometric curvature minima that organize expansion-dominated cosmic voids, complementary to black holes and other mass-concentration nodes that organize matter-dominated structures. Together these define Curvature Extremum Duality: a dual organizing framework in which spacetime structure arises around both maxima and minima of curvature.

A direct observational discriminator of curvature-minimum (void-node) geometry follows immediately: at fixed, independently measured distance, redshift residuals should correlate with the integrated void-node geometry along the line of sight. Lines of sight traversing deeper or more extended void-node interiors should exhibit systematic redshift excess relative to comparably distant sources viewed through less void-dominated geometry.

Redshift emerges naturally as a clean and testable observable consequence of this geometry, encoding real affine path length through void-organized spacetime rather than purely kinematic recession. Whether MDNs represent a fundamental organizing principle of cosmological geometry or a powerful interpretive lens will be decided empirically. Either way, treating curvature minima as active geometric participants rather than passive absence provides a richer and more faithful description of the universe we observe.

Formal relativistic optical derivations and explicit metric realizations are developed in companion papers.

Appendix A

Heuristic Toy Model for Path-Integrated Geometric Redshift in MDN-Organized Void Geometry

A.1 Purpose and Status of This Appendix

This appendix presents a heuristic toy model designed to demonstrate that path-integrated geometric redshift accumulated during photon propagation through MDN-organized (void-node-organized) void geometry is:

- correct in sign,
- plausible in order of magnitude, and
- observationally testable with existing data.

The model is intentionally simplified and serves a plausibility-establishing role, not a full theoretical derivation. In particular:

- the redshift accumulation law is adopted phenomenologically,
- voids are treated as idealized spherical regions with uniform underdensity,
- relativistic gauge subtleties are not resolved,
- and no explicit spacetime metric is constructed.

Numerical estimates should therefore be interpreted as conservative lower bounds.

The goal is to show that the curvature-extremum MDN framework, in which MDNs act as curvature minima complementary to mass-concentration nodes, leads naturally to quantitative effects of potentially observable size, motivating more rigorous metric-level development.

A.2 Distinction from Potential-Based Redshift Effects

The redshift mechanism considered here is categorically distinct from the Integrated Sachs–Wolfe (ISW) and Rees–Sciama effects.

ISW and Rees–Sciama redshifts arise from time-varying gravitational potentials. In static or slowly evolving potentials, photon energy shifts on entry and exit largely cancel, leaving only small residual effects proportional to the temporal evolution of the potential.

By contrast, the MDN framework concerns adiabatic wavelength stretching driven by spacetime geometry itself, not by exchange of energy with a gravitational potential. The effect accumulates continuously along the null geodesic as the photon propagates through expansion-dominated geometry. When the photon exits the void region, it carries the stretched waveform forward; there is no reciprocal "exit" process that reverses the deformation. This is consistent with MDNs being geometrically complementary curvature extrema rather than inverse potential wells.

As a result, MDN geometric redshift is path-integrated, not potential-integrated, and does not rely on temporal evolution of the potential during traversal. This distinction explains why cancellation arguments applicable to ISW or Rees–Sciama effects do not apply here.

A.3 Heuristic Redshift Accumulation Law

We adopt the phenomenological relation

$$dz = \frac{H(l)}{c}\, dl,$$

where $H(l)$is the local effective expansion rate encountered along the photon path.

This expression is intended as a heuristic representation of cumulative adiabatic stretching along a null geodesic in expansion-dominated regions. A fully covariant derivation requires explicit specification of the spacetime metric and is deferred to future work.

A.4 Local Expansion Rate Inside a Void

Consider a void of effective radius R (organized around an MDN) with uniform matter underdensity $\delta < 0$, embedded in a background FRW cosmology.

At low redshift, the local expansion rate inside the void may be approximated using the Friedmann equation:

$$H_{\text{void}}^2 = H_0^2[\Omega_m(1+\delta) + \Omega_\Lambda].$$

For illustrative purposes, a conservative linear approximation yields the excess expansion rate

$$\Delta H \approx \frac{1}{2} H_0\, \Omega_m \mid \delta \mid.$$

For transparency, we adopt a matter-dominated approximation to isolate the contribution arising from reduced gravitational resistance inside underdense regions. Including the cosmological constant reduces the fractional contrast in the local expansion rate but does not alter the scaling behavior or sign of the effect. The approximation therefore serves as a controlled, order-of-magnitude estimate rather than a fine-tuned maximization of redshift accumulation.

A.5 Path Length Through Void Geometry

In this formulation, geometric redshift accumulation depends only on the amount of expansion-dominated geometry traversed, not on any directional projection of expansion.

For a spherical void, the photon travels a chord length determined by its impact parameter $b$relative to the void center:

$$L_{\text{void}}(b) \;=\; 2\sqrt{R^2 - b^2}(b < R),$$

$$L_{\text{void}} = 0(b \geq R).$$

This formulation reflects the volumetric nature of void expansion organized around a curvature minimum (MDN): the local expansion rate $H_{\text{void}}$is a scalar property of the region, and redshift accumulation depends on how long the photon dwells within that region.

A.6 Single-Void Excess Redshift

The excess redshift accumulated during traversal of a single void is therefore

$$\Delta z_{\text{void}} \;=\; \frac{\Delta H}{c}\, L_{\text{void}}(b).$$

This expression has the correct limiting behavior:

- it is maximized for central paths ($b = 0$),
- it vanishes only when the path length through the void goes to zero,
- and it does not require orientation-dependent projection factors.

A.7 Numerical Estimates for a Representative Supervoid

Using conservative parameters typical of large observed supervoids:

- $H_0 = 70\ \mathrm{km\ s^{-1}\ Mpc^{-1}}$,
- $\Omega_m = 0.30$,
- $|\ \delta\ | = 0.3$–0.8,
- $R = 150$–$300\ \mathrm{Mpc}$,

one finds

$$\Delta z_{\mathrm{void}}\ \sim\ 10^{-3}\text{–}10^{-2},$$

depending on void size and underdensity.

Such redshift excesses are modest for individual voids but are detectable for ensembles of objects with independent distance estimates (e.g., Type Ia supernovae).

A.8 Multi-Void Accumulation

Cosmological lines of sight traverse a connected network of void regions organized around multiple MDNs (void nodes). For Nsubcritical void crossings,

$$\Delta z_{\mathrm{total}}\ =\ \sum_{i=1}^{N} \Delta\, z_i.$$

Under the approximation of statistically uniform void properties, this leads to

$$\Delta z_{\mathrm{total}} = \frac{H_0\, \Omega_m\ \mid \delta_{\mathrm{mean}} \mid\ f_{\mathrm{void}}}{3c}\ D,$$

where:

- $f_{\mathrm{void}}$is the void filling fraction by volume,
- $D$is the comoving distance.

The cumulative effect scales linearly with distance, producing a Hubble-like contribution to the observed redshift.

A.9 Implications for the Hubble Tension

For representative values

- $|\delta_{\text{mean}}| = 0.3\text{–}0.5$,
- $f_{\text{void}} \approx 0.7\text{–}0.8$,

the effective excess expansion rate is

$$\Delta H_{\text{MDN}} \sim 1.5\text{–}3\ \text{km s}^{-1}\ \text{Mpc}^{-1},$$

accounting for roughly 20–50 % of the observed discrepancy between local and early-universe determinations of H_0.

This result does not claim to replace cosmic expansion or dark energy. Instead, it suggests that line-of-sight geometric bias may contribute systematically to locally inferred expansion rates. This feature arises naturally from curvature-minimum geometry that is complementary to curvature-maximum structures.

A.10 Observational Test

The MDN framework predicts that apparent redshift residuals and inferred H_0 values should correlate with the integrated void geometry along each line of sight.

Define the observable statistic

$$\Xi_{\text{MDN}} = \sum_i |\delta_i|\ L_{\text{void},i}(b_i),$$

where the sum runs over voids intersected by the line of sight.

This statistic weights line-of-sight contributions by the geometric influence of curvature-minimum organizing nodes, not merely by density contrast.

A positive correlation between $\Xi_{\mathrm{MDN}}$and supernova H_0residuals constitutes a direct, falsifiable test using existing datasets (e.g., Pantheon+) and current void catalogs.

A.11 Limitations and Directions for Further Work

The present model is limited by:

- its heuristic redshift accumulation law,
- idealized void geometry,
- absence of an explicit spacetime metric,
- and unresolved gauge considerations.

A rigorous treatment requires formulation within an inhomogeneous metric that encodes curvature minima as organizing nodes under curvature extremum duality, rather than treating voids solely as density perturbations. This step is reserved for future work.

A.12 Summary

This appendix demonstrates that path-integrated geometric redshift in MDN-organized void geometry is:

- conceptually well motivated within the framework of curvature extremum duality,
- quantitatively plausible at observable levels,
- and directly testable.

Because cosmological parameters used here are inferred under the assumption that redshift is purely kinematic, the numerical estimates should be interpreted as conservative lower bounds on the MDN contribution.

The model is deliberately conservative and incomplete, yet sufficient to motivate deeper mathematical and observational investigation of MDN-based geometric effects in cosmology.

The appendix establishes plausibility of observable effects arising from curvature-minimum (MDN/Void-node) geometry, complementary to the well-studied curvature-maximum phenomena associated with black holes.